

for a given security,” these amendments will “significantly increase the capacity requirements for market participants to process these additional quotation messages.”¹³²⁴ Specifically the commenters stated that the additional message traffic could impact CAT costs which they state was not analyzed by the Commission.¹³²⁵

Commenters also stated that increased message traffic can increase market participants’ technological needs in terms of computing and processing requirements.¹³²⁶ Some commenters stated that more message traffic could slow down markets as it would take more time for market participants to process the increased data, this could in turn disrupt trading strategies.¹³²⁷ Another commenter stated that increased message traffic would lead to increased technology costs in terms of server, processor, and storage requirements needed to deal with increased message traffic.¹³²⁸ One commenter stated that significant increases in message traffic can cause technological difficulties for the exchanges.¹³²⁹ Another commenter stated that increased message traffic would give an advantage to sophisticated algorithmic traders who have a greater capacity to manage the cost of the data and could better apply the data itself.¹³³⁰

The Commission recognizes the potential for these costs articulated by the commenters, and considers the information provided. However, the Commission expects these effects, including the effect on CAT costs, to be mild. This conclusion stems from experience in options

¹³²⁴ See SIFMA Letter II at 21, stating that it is concerned about the impact of the amendments on “consolidated market data, capacity costs and CAT data requirements.”

¹³²⁵ See SIFMA Letter II at 21, 43; see also Citadel Letter II at 4-5 and FIF Letter at 10-12.

¹³²⁶ See, e.g., BlackRock Letter at 5, GTS Letter at 5, FIF Letter at 1, 6-8, RBC Letter at 4, IBKR Letter at 5, Fidelity Letter at 17.

¹³²⁷ See Tradeweb Letter at 2.

¹³²⁸ See STA Letter at 6.

¹³²⁹ See Budish Letter at 5.

¹³³⁰ See, e.g., Goldman Sachs Letter at 8 and RBC Letter at 4.

markets, which, as discussed above, experience considerably more message traffic than equity markets without significant adverse effects.¹³³¹ It also stems from research provided by commenters that gives a framework for estimating the increase in message traffic and the costs of such an increase which will now be discussed suggesting an estimated overall increase in technology costs to market participants of approximately 1%, and an estimated increase in CAT costs of approximately \$4.1 million.

The amendments add one additional tick intra-spread for some stocks. The Commission expects the increase in message traffic from this increase to be mild. One commenter referenced research performed by one of the exchanges suggesting that each additional tick intra-spread adds 20% to message traffic.¹³³² This research suggests message traffic could increase 20% for stocks receiving the smaller tick size. The Commission expects 74% of share volume to receive the smaller tick, and the remaining share volume to be unchanged by the amendments. Multiplying 74% by 20% suggests that, based on the information provided by commenter, it is reasonable to believe total message traffic may increase by about 15% due to the amendments.

Accordingly, the costs associated with this increased message traffic on most market participants are expected to be relatively small. One commenter estimated that the proposed rule would lead to an increase in total infrastructure costs of 10% due to increased message traffic.¹³³³

¹³³¹ See discussion surrounding supra notes 239 to 250.

¹³³² See FIF Letter at 7 referencing Phil Mackintosh, More Ticks, More Messages (Oct. 27, 2022), available at <https://www.nasdaq.com/articles/more-ticks-more-messages>. The research provided by the exchange provided three data points, the baseline case with no increase in ticks, a 1:5 increase and a 1:10 increase which are associated with 0%, 100% and 500% increase in message traffic. If message traffic is the variable on the vertical axis and the tick size change the horizontal axis, then all three of these data points fall on the line % Increase in Message Traffic = $0.2 * \text{Ticks Added Intra Spread}$. Substituting in the number 1, to indicate the number of ticks added intra-spread yields an increase in message traffic of 1.2, or a 20% increase.

¹³³³ See Fidelity Letter at 17.

Using the above methodology we estimate that if the proposal would have led to an increase in message traffic related costs of 10%, as suggested by the commenter, and if the costs of message traffic are proportional to the increase in message traffic, then by this reasoning, the adopted amendments, which create fewer ticks, will be associated with an increased infrastructure costs of approximately 1%.¹³³⁴ Additionally, another exchange, commenting on the Proposing Release which had more and smaller ticks than the adopted amendments, estimated that the increase in message traffic due to the proposal would be relatively small compared to historical variation.¹³³⁵

Commenters asked the Commission to estimate the effect of the tick size reduction on CAT costs via the mechanism of increased message traffic.¹³³⁶ The fourth quarter 2023 CAT cloud hosting services costs were \$30.47 million,¹³³⁷ or \$121.88 million if annualized. Further, a recent order approving an amendment to the NMS plan governing CAT stated that equity message traffic accounted for 23% of total CAT message traffic (with options traffic accounting

¹³³⁴ The proposal would have added 9 ticks for the stocks receiving the \$0.001 tick size, 4 ticks for the stocks receiving the \$0.002 tick size, and 1 additional tick for the stocks receiving the 0.005 tick. Using the methodology provided by the commenters suggesting that each additional tick would increase message traffic 20% relative to the baseline suggests that stocks receiving the \$0.001 tick would have an estimated message traffic increase of 180%, for stocks receiving the \$0.002 tick the estimated increase in message traffic would be 80%, and for stocks receiving that \$0.005 tick the increase in message traffic would be 20%. Using the same methodology as used to create Table 7 to estimate the amount of trading volume that would have been associated with each of the proposed tick sizes based on 2023 data suggests that approximately 50% of trading volume would have received the \$0.001 tick, approximately 25% of trading volume would have been associated with the \$0.002. Combining these figures suggests that message traffic associated with the proposal would have increased approximately 125%. Thus, if a 125% increase in message traffic is expected to be associated with a 10% increase in infrastructure costs, then assuming that costs are proportional to message traffic, then a 15% increase in message traffic will be associated with just over a 1% increase in infrastructure costs associated with the amended Rule.

¹³³⁵ See NYSE Letter II at 10-11.

¹³³⁶ See FIF Letter at 10-12, SIFMA Letter II at 43.

¹³³⁷ See Consolidated Audit Trail, LLC, 2023 Financial and Operating Budget, [available at](https://www.catnmsplan.com/sites/default/files/2024-01/01.17.24-CAT-Q4-2023-Budget-vs-Actual.pdf) <https://www.catnmsplan.com/sites/default/files/2024-01/01.17.24-CAT-Q4-2023-Budget-vs-Actual.pdf>.

for the remainder).¹³³⁸ Thus, estimated CAT costs associated with equity message traffic equal \$121.88 million*23% = \$28.03 million. Under the assumption that CAT costs are linear in message traffic,¹³³⁹ an increase in equity message traffic of 15% would correspond to an increase in CAT costs according to the following formula: \$28.03 million *15%(expected increase in message traffic) ≈\$4.1 million. Based on this data, we estimate CAT costs would increase by an estimated \$4.1 million per year due to increased equity message traffic associated with the smaller tick size. The CAT NMS Plan requires participants and industry members to fund the CAT. The funding model for the CAT NMS Plan allocates the costs to fund CAT among participants and the members of a national securities exchange or a member of a national securities association.¹³⁴⁰

Commenters stated that a smaller tick size will fragment liquidity in the order book and reduce the displayed liquidity at the NBBO, which in turn reduces the information about liquidity available in the market for market participants who do not receive depth of book

¹³³⁸ See SEC, Joint Industry Plan; Order Approving an Amendment to the National Market System Plan Governing the Consolidated Audit Trail; Notice (Joint Industry Plan), Securities Exchange Act Release No. 98290 (Sep. 6, 2023), 88 FR 62628 (Sep. 12, 2023) at 62680 & n.1077.

¹³³⁹ The relation between total CAT cost and volume is complex and driven by both volume and the complexity of processing the additional information and may not always be linear. See Letter from Brandon Becker, Chair, CAT NMS Plan Operating Comm., to Vanessa Countryman, Sec’y, SEC at 3 n.11 (Mar. 27, 2024), available at <https://catnmsplan.com/sites/default/files/2024-03/03.27.24-Proposed-CAT-NMS-Plan-Amendment-Cost-Savings-Amendment.pdf>. However, one commenter speaking generically about server and technology cost associated with message traffic stated that these costs were “proportional to the increase in message traffic.” See FIF Letter at 9. Consequently, there is uncertainty regarding the exact relation between CAT costs and equity message traffic. However, to facilitate estimates of the effect of the amendments on CAT costs, the Commission makes the assumption that CAT costs are linear in message traffic – knowing that there is uncertainty regarding the exact relation. However, without some assumption about the relation between CAT costs and message traffic, no inference could be made. To the extent that CAT costs are convex or concave – rather than linear - in message traffic then the actual costs associated with the amendments would be higher or lower than what is estimated here.

¹³⁴⁰ See Joint Industry Plan, supra note 1338.

information from proprietary data feeds.¹³⁴¹ These commenters stated that this could increase reliance on and the subsequent cost of proprietary data products due to the increased need for the data and simply due to additional data points.

The Commission acknowledges that, in addition to lowering transaction costs, the amendments will likely also spread depth across more price points in the limit order book and reduce depth at the NBBO in some stocks with a 0.5 cent tick, which could increase the complexity and cost associated with sourcing liquidity for larger orders.¹³⁴² However, the inclusion of odd-lot information in consolidated market data is anticipated to help to mitigate this effect, and the eventual inclusion of depth of book information in consolidated market data due to the implementation of the MDI Rules should also help mitigate this effect.¹³⁴³ Fragmentation of liquidity in the limit order book will increase the precision and usefulness of information about depth of book quotes and odd-lot quotes inside the NBBO for some stocks with a 0.5 cent tick because orders in these stocks will be displayed at half-penny increments rather than at penny increments, which provides more precise information.¹³⁴⁴ This will increase demand for depth of book data, which could cause more market participants that currently only subscribe to

¹³⁴¹ See Citadel Letter I at 8, Virtu Letter II at 8, 10, Citigroup Letter at 4, SIFMA Letter II at 41-42, and Lewis Letter attached to Virtu Letter II at p 35. See also Lewis Letter attached to Virtu Letter II at 33 (pointing out that the need to manage increased message traffic could lead proprietary data feed to become more expensive independent of market participant reliance such data products). SRO fees are subject to the rule filing process under Exchange Act section 19(b) and Rule 19b-4. 15 U.S.C. 78s(b); 17 CFR 240.19b-4.

¹³⁴² See supra section VII.D.1.b.i (discussing the smaller tick size fragmenting depth across more price levels).

¹³⁴³ After the MDI Rules are implemented, consolidated market data is expected to contain depth information for five price levels beyond the NBBO, which will help mitigate the effects of a reduction in displayed depth at the NBBO from a reduction in tick size. See MDI Adopting Release, supra note 10, at 18625-30 and 18755-59 for further discussions on the content and effects of MDI depth of book data.

¹³⁴⁴ This could increase the usefulness of both exchange proprietary depth of book feeds and MDI odd-lot information and depth of book data relative to SIP data for stocks with a 0.5 cent tick. However, as discussed below, the usefulness of MDI depth of book data in comparison to exchange proprietary depth of book data may decrease for these stocks.

SIP data to purchase exchange proprietary depth of book feeds. It could also cause some market participants that only subscribe to SIP data and who previously would not have purchased MDI depth of book data or odd-lot information to decide to purchase one or both of these MDI data elements when they become available. The Commission acknowledges that the amendments could increase the cost associated with producing exchange proprietary data feeds and MDI data, but the cost increases may not be significant.¹³⁴⁵ Any changes in the prices of exchange proprietary data feeds would be subject to the SRO fee filing process.¹³⁴⁶

After the MDI Rules are fully implemented, commenters stated that the amendments could decrease the benefits of MDI data, which will include depth of book data for prices with quotes that are five levels outside of the NBBO.¹³⁴⁷ The tick reduction will likely alter the information in MDI data for some stocks with a 0.5 cent tick, making MDI depth of book data less informative while increasing the informativeness of MDI odd-lot information. The tick reduction may generally result in MDI depth of book data showing information on fewer shares in the order book for some stocks with a 0.5 cent tick. More specifically, it may no longer include some information on orders that are \$0.03 to \$0.05 away from the NBBO in some stocks with a 0.5 cent tick.¹³⁴⁸ However, the effects of the loss of this information may not be

¹³⁴⁵ See supra note 1334 and accompanying text (estimating increased infrastructure costs of approximately 1% due to an increase in message traffic).

¹³⁴⁶ SRO fees are subject to the rule filing process under Section 19(b) and Rule 19b-4. 15 U.S.C. 78s(b); 17 CFR 240.19b-4.

¹³⁴⁷ SIFMA Letter II at 41-42 stated that the Rule, by making NMS data under MDI less competitive with proprietary feeds, will offset some of the benefits of the MDI Rules.

¹³⁴⁸ The MDI NMS data includes information on five round lot price levels outside the NBBO. The tick reduction implies that the levels may be \$0.005 apart from each other; without the tick reduction, the levels would be at least \$0.01 apart. Therefore, the tick reduction may result in five levels that cover orders that are \$0.025 outside the NBBO, whereas without the tick reduction the five levels would cover orders that are \$0.05 outside the NBBO. It is possible that MDI NMS data will include depth that is greater than \$0.025 outside the NBBO for stocks receiving a tick reduction—this is because each displayed level is

significant because the MDI depth of book data would still show five round lot price levels of depth and many market participants may not need more than five levels of depth of book information.¹³⁴⁹ Additionally, for stocks with a 0.5 cent tick, information about depth within \$0.025 of the NBBO may be more valuable than information about depth that is deeper in the book (e.g., between \$0.03 and \$0.05 of the NBBO), because only a relatively small portion of marketable orders execute at prices outside the NBBO.¹³⁵⁰ Further, for stocks that have the \$0.005 tick, MDI odd-lot information about odd-lots inside the NBBO will likely be more valuable because it will show these odd-lots on a finer grid—as long as the NBBO spread is \$0.01 or more, the tick reduction allows investors using MDI NMS data to see odd-lots at finer half-penny increments. For stocks that retain the one cent tick size, the tick size amendments are unlikely to affect the informativeness of MDI depth of book data or odd-lot information because the amendments are unlikely to change the trading environment, including odd-lot quotes inside the NBBO and depth in the limit order book.¹³⁵¹

Commenters stated that reducing the tick size would make MDI data less competitive with exchange proprietary data feeds.¹³⁵² The tick reduction may cause some market participants that currently utilize exchange proprietary depth of book feeds, and would have replaced them

required to comprise at least one round lot worth of shares, and odd-lot orders will be aggregated with other orders to form a level with a sufficient number of shares (with the displayed price equaling the least competitive price among the aggregated orders). That is, if there is not enough liquidity at a given price point within \$0.025 of the NBBO, then a price point further outside the NBBO will be displayed.

¹³⁴⁹ See MDI Adopting Release, supra note 10, at 18625 nn.387, 388 (discussing levels of depth used in order routing and analysis finding that a significant percentage of the total notional value of all depth of book quotations for both liquid and illiquid stocks falls within the first five price levels).

¹³⁵⁰ See MDI Adopting Release, supra note 10, at 18731 (discussing analysis showing that only a small percentage of orders execute outside the NBBO).

¹³⁵¹ See supra section VII.D.1.

¹³⁵² See, e.g., SIFMA Letter II at 41-42 and Citadel Letter I at 8.

with MDI depth of book data and odd-lot information if it was available at a lower cost,¹³⁵³ to no longer substitute MDI depth of book data and odd-lot information for proprietary feeds.¹³⁵⁴

Proprietary feeds will likely gain certain advantages over MDI depth of book data as a result of the tick reduction. First, self-aggregators and competing consolidators relying on MDI depth of book data may not be able to provide information on depth for orders that are between \$0.03 and \$0.05 away from the NBBO for some stocks that have been assigned a \$0.005 tick size because the MDI depth of book data only provides data on depth at 5 round lot price levels. Second, in cases where there is insufficient liquidity at a given price level to form a round lot due to the reduction in the tick spreading liquidity out across more levels, self-aggregators and competing consolidators relying on MDI depth of book data will not be able to provide information on depth at each price point beyond the NBBO, but will rather only have information on liquidity that is aggregated over multiple price points to compose a round lot; proprietary feeds, on the other hand, can offer information on available liquidity at each price point. These two advantages may lead some market participants that would have substituted MDI depth of book data for exchange proprietary depth of book data, if it was available at a lower cost once it was implemented, not to do so due to the tick reduction. To the extent this occurs, it would result in a transfer from competing consolidators and the market participants who would have substituted

¹³⁵³ In the MDI Adopting Release, the Commission stated that it believes that the total fees for the equivalent of consolidated market data are likely to decline because of the MDI Rules, but recognizes uncertainty about how the effective national market system plan(s) will set the fees for data content underlying consolidated market data offerings and how SROs will set the fees for connectivity necessary to receive the data content underlying consolidated market data as well as how the competing consolidators will price their services. See MDI Adopting Release, supra note 10, at 18772.

¹³⁵⁴ This would apply to both market participants that would have self-aggregated the MDI depth of book data and odd-lot information or purchased it from a competing consolidator. See MDI Adopting Release, supra note 10, at 18793-95 for discussions on market participants substituting consolidated market data for exchange proprietary feeds. However, as discussed above, demand for MDI depth of book data could increase for market participants that currently rely on SIP data.

MDI depth of book data for their proprietary feeds to exchanges, because these market participants will continue paying, and exchanges will continue to receive revenue, for their proprietary data feeds.¹³⁵⁵ On the other hand, to the extent that the tick size reduction increases the usefulness and demand for depth of book data—as discussed above—some market participants that currently rely on SIP data may be able to satisfy their increased data needs by purchasing MDI odd-lot information and depth of book data from a competing consolidator once the MDI rules are implemented.¹³⁵⁶

d. Analysis of the Length of Evaluation and Operative Periods

The adopted amendments specify three-time periods that govern tick assignment.¹³⁵⁷ First are the periods that a new tick size is operative. The adopted rules update the tick size twice a year, so they set this period at six months. Second, the rules set the ticks based on two backward-looking three-month Evaluation Periods over which NMS stocks' TWAQS is measured; if the TWAQS is at or below \$0.015 during this Evaluation Period, then the stock will be assigned a half-penny tick. Otherwise, it will receive a penny tick. Finally, there is a one-month gap between the Evaluation Period and the initiation of the subsequent tick size. The January through March Evaluation Period will determine the tick for the May through October operative period.

¹³⁵⁵ See MDI Adopting Release, *supra* note 10, at 18793-95 for discussions on market participants substituting consolidated market data for exchange proprietary feeds and related transfers.

¹³⁵⁶ See *infra* section VII.D.4.b for additional discussions of why low latency traders may not substitute odd-lot information from the exclusive SIPs for exchange proprietary feeds.

¹³⁵⁷ See 17 CFR 242.612(a)(1), defining the evaluation period to assign a tick size as “(i) the three months from January through March of a calendar year and (ii) the three months from July through September of a calendar year during which the Time Weighted Average Quoted Spread of an NMS stock shall be measured by the primary listing exchange to determine the minimum pricing increment for each NMS stock,” and 17 CFR 242.612(b)(1), and specifying the pricing increment will be operative “(a) the first business day of May for the Evaluation Period from January through March and continue through the last business day of October of the calendar year, and (b) the first business day of November for the Evaluation Period from July through September and continue through the last business day of April of the next calendar year.”

Likewise, the July through September Evaluation Period will determine the tick for November through April of the subsequent calendar year.

Commenters highlighted a tradeoff with respect to the proposed backward-looking evaluation period. On one hand, a short period “uses the period immediately closest in time to measure securities’ behavior to ensure that tick-constrained names are selected using the most recent and relevant basis...”¹³⁵⁸ On the other hand, a short period may place too much emphasis on idiosyncratic and unrepresentative events.¹³⁵⁹

Commenters also highlighted a tradeoff regarding the proposed length of the operative period. Frequent updating will allow tick assignment to more quickly react to changes in the “market environment and individual stock behavior...”;¹³⁶⁰ quicker updating would thereby “reduce how often and for how long a stock’s tick size stays outside the optimal range.”¹³⁶¹ However, more frequent updating may impose costs in terms of adjustments to algorithms, operations, trading models and systems,¹³⁶² customer complaints,¹³⁶³ and tick size oscillation.¹³⁶⁴

Commenters also discussed the importance of a gap between the proposed evaluation period and the beginning of the subsequent tick size to give industry time to update systems and avoid disruptions.¹³⁶⁵ The proposal did not include a gap between the evaluation period and the

¹³⁵⁸ See NYSE Letter I at 4.

¹³⁵⁹ See FIA PTG Letter II at 2, Optiver Letter at 2, and MMI Letter at 6.

¹³⁶⁰ See NYSE Letter I at 4.

¹³⁶¹ See Pragma Letter at 6.

¹³⁶² See FIA PTG Letter II at 2, UBS Letter at 13, Citigroup Letter at 2, BlackRock Letter at 9, Hudson River Letter at 3, JPMorgan Letter at 4, Morgan Stanley Letter at 3, State Street Letter at 3, and MMI Letter at 6.

¹³⁶³ See TradeStation Letter at 5.

¹³⁶⁴ See Mitre Corp. Letter at 5 and MEMX Letter at 16.

¹³⁶⁵ See RBC Letter at 3, T. Rowe Price Letter at 4, and IEX Letter I at 7. T. Rowe Price and IEX both suggested a 1-month gap. RBC suggested “an appropriate amount of time to be informed of any changes in order to implement them, and to minimize errors as much as possible.”